Supporting figures and prespecified analyses

Individual aversive emotional reactivity in HABS

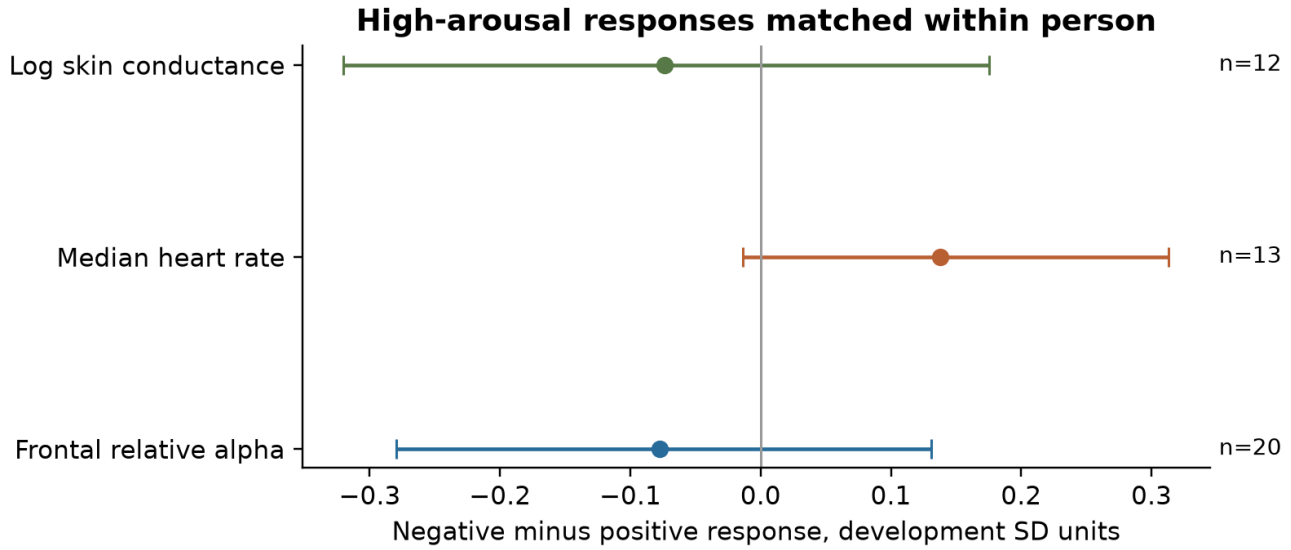


Figure 3. Within-identifier negative versus positive valence at arousal >=6, matched without reuse within 0.5 arousal units. Physiology is centered on development video means and each person's J1, with development-only scales. Intervals resample identifiers; marker support differs with sensor availability. This descriptive comparison does not validate stress specificity.

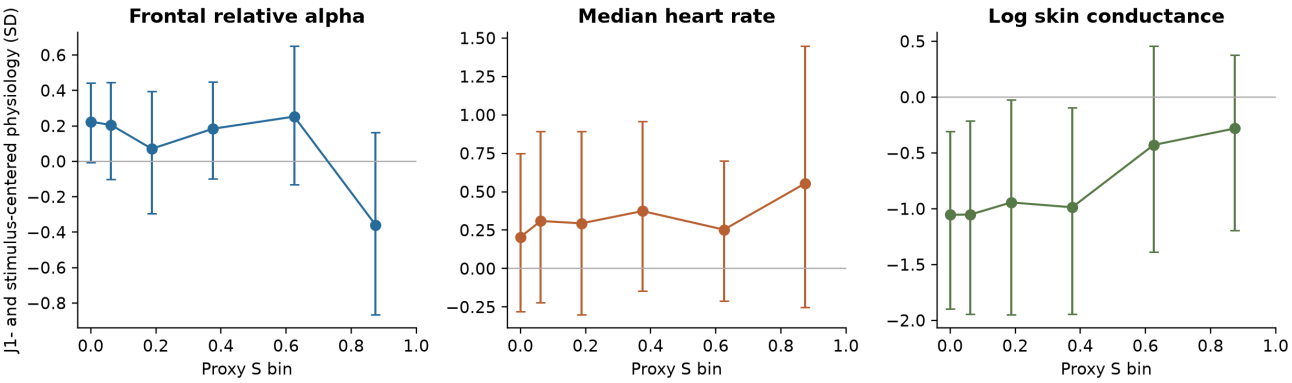


Figure 4. Fixed score bins and identifier-balanced physiological means, after development-video and J1 centering. The first bin contains exact zeros. Points require at least five contributing identifiers; confidence intervals are descriptive and uncorrected. Curvature may reflect the constructed target and does not establish a biological threshold.

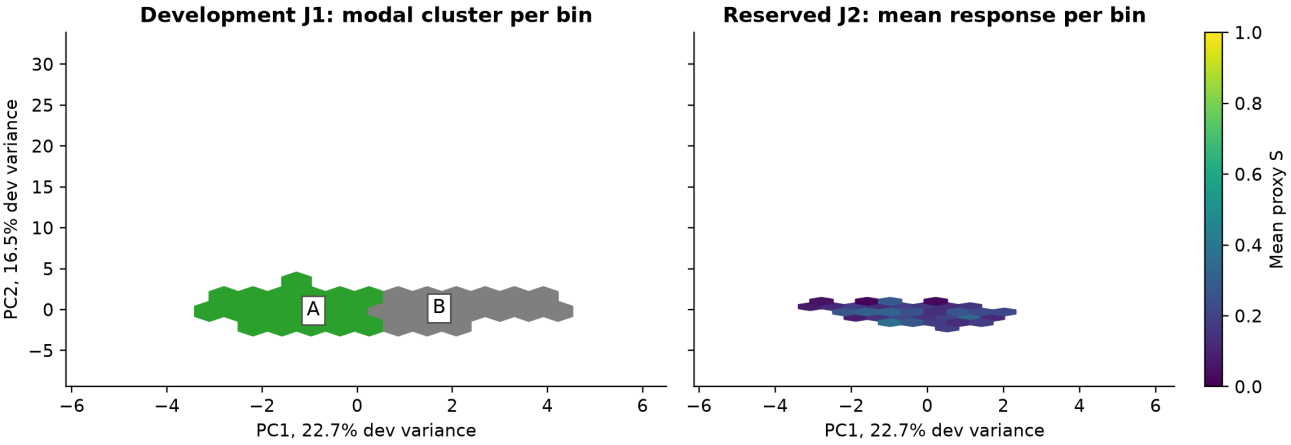


Figure 5. Complete-case development J1 physiology after development-video centering. Clusters and the PCA projection were fitted without response labels. The left panel shows the modal cluster in bins; the right shows mean reserved J2 score after learning. Bins require at least five observations. Axes retain the range of outlying development coordinates even when low-count cells are not displayed. A two-dimensional projection alone is not evidence of discrete biological populations.

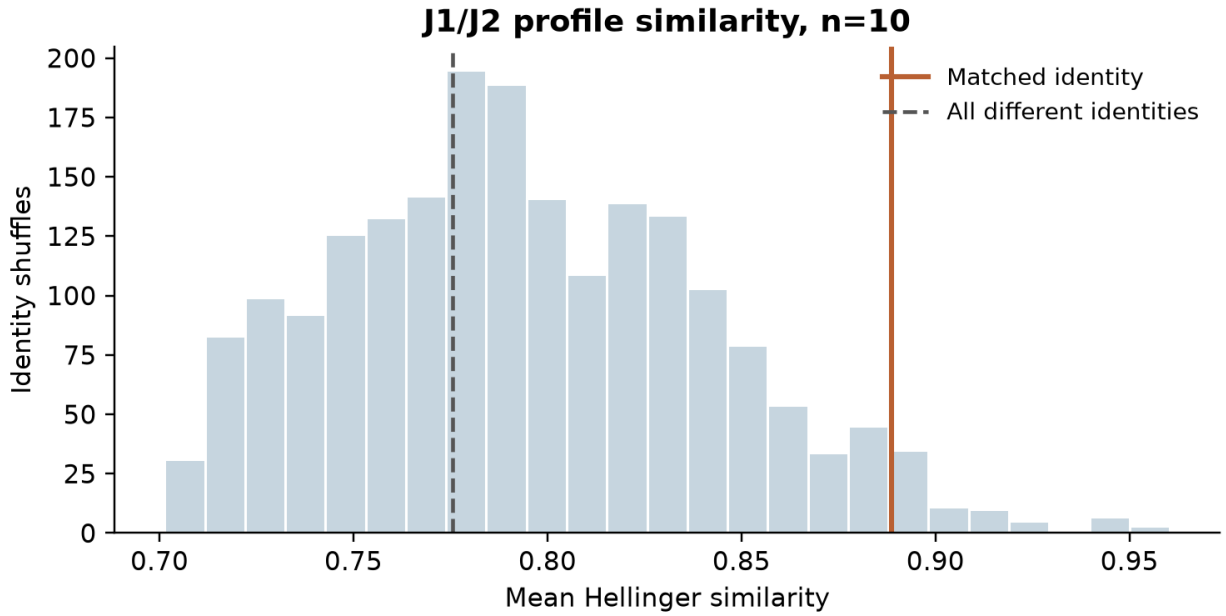


Figure 6. Similarity of a person's J1/J2 cluster-frequency descriptors compared with different identities and count-stratified identity shuffles. This observational reference preserves trial-count strata but cannot establish stress-specific or artifact-free traits. The predictive comparison M5 versus M4 and continuous-history control remains the utility test.

Secondary endpoints

Analysis	Incremental gain and 95% interval
valence (rating units)	-0.0008 [95% CI -0.0171, +0.0156]
arousal (rating units)	+0.0037 [95% CI -0.0140, +0.0223]
alternative	-0.0010 [95% CI -0.0059, +0.0035]
negative	-0.0004 [95% CI -0.0027, +0.0017]

The negative-valence endpoint conditions on current reported arousal and is explanatory. The alternative target uses the square root of normalized activation. All selections use development only. Secondary intervals are exploratory; no multiplicity-adjusted discovery claim is made.

Extreme-trained score and intermediate responses

The high threshold was fixed from development at S=0.56908. Supervised fitting used 3,321 zero/high observations; 380 reserved intermediate trials were excluded from that supervised fit. Mean within-identifier rank correlation on intermediate responses was 0.295 [95% CI 0.132, 0.441] for stimulus/context alone and 0.340 [0.180, 0.484] for the physiological score, with 18 contributing identifiers. A gradient is therefore present without physiology. These overlapping descriptive intervals do not establish a paired physiological advantage, and neither score is a calibrated quantity of stress.